Chapter 2: Variables and Data Types

Python Programming Study Guide Series

VARIABLES IN PYTHON

Variables serve as named containers that hold references to memory locations containing data. Assignment is performed using the single equal operator (=).

```
# Variable Assignment age = 25 price = 19.99 user_name = "Alice" is_active = True
```

STANDARD DATA TYPES

Type	Class Name	Example	Description
Integer	`int`	`x = 100`	Whole numbers of arbitrary length
Float	`float`	`g = 9.81`	Floating-point real numbers
String	`str`	`s = "Python"`	Sequence of Unicode characters
Boolean	`bool`	`flag = False`	Logical truth values (`True` or `False`)
List	`list`	`items = [1, 2]`	Mutable ordered sequence
Dictionary	`dict`	`d = {"a": 1}`	Key-value mappings

Naming Conventions & Rules

- Variable names can only contain alphanumeric characters and underscores (`a-z`, `A-Z`, `0-9`, and _).
- Names must begin with a letter or an underscore, never a number.
- Use **snake_case** for variable names (e.g., `total_amount`).